Shri Sant Gajanan Maharaj College of Engineering Shegaon
Department of Information Technology
Session 2024-2025

Report on Innovative Teaching-Learning Practice: Virtual Lab for DSS
Name of the Faculty: Anand G Sharma
Designation: Assistant Professor
Department: Information Technology
Date: 16th Nov. 2024

---

1. Introduction

For subject Data Science and statistics python is favored programming language, as a part of innovative teaching-learning practices for enhancing student engagement, conceptual understanding, and skill development. Among these, **Virtual Labs** have emerged as an effective tool to bridge the gap between theoretical knowledge and practical application. Virtual labs provide a simulated environment where students can perform experiments, visualize outcomes, and understand concepts without being restricted by physical resources or laboratory availability.

---

2. Objectives of Conducting Virtual Lab

- To provide students with hands-on experimental learning in a safe and controlled environment.
- To enhance conceptual clarity by allowing repeated trials and instant feedback.
- To integrate technology into pedagogy, promoting self-paced and experiential learning.

---

3. Methodology

- The session was conducted using on IITB Virtual Lab of python Programming link: https://python-iitk.vlabs.ac.in/List%20of%20experiments.html
- Students were introduced to the interface and guided on how to navigate and perform experiments virtually.
- Selected experiments relevant to the course curriculum were demonstrated, after which students performed them individually.
- A post-lab discussion was carried out to analyze the results and relate them to real-world applications.

---

4. Outcomes and Benefits

- **Active Engagement:** Students were more interactive and motivated to explore beyond the syllabus.
- **Flexibility:** Experiments could be repeated multiple times, strengthening learning without additional cost.
- **Accessibility:** Even students with limited access to campus facilities could participate remotely.
- **Skill Development:** Improved digital literacy, problem-solving ability, and independent learning.
- **Integration with Theory:** Concepts discussed in lectures were reinforced with practical virtual demonstrations.

---

5. Photograph of Event

Fig: Students Conducting Virtual Lab session on Python Programming

6. Attendance report

IIT Bombay | Virtual Labs

Attendance Report - 2024 Date:16-11-2024

Institute : SHRI SANT GAJANAN MAHARAJ COLLEGE OF ENGINEERING, SHEGAON

NCID : 64

Roll No	Name	Department
307221	Aditya Ranmale	Information Technology
-	Ambika Patil	Information Technology
307044	Ameya Mandwale	Information Technology
03	APURVA MASNE	Information Technology
307345	Atharva Harsulkar	Information Technology
307148	ayush kahar	Information Technology
4	Bhakti Jasani	Information Technology
307450	Gauri Khandar	Information Technology
8	Gauri Sambare	Information Technology
12	Isha Wagh	Information Technology
307503	Ishwar Sharma	Information Technology
307309	KRISHNA BHISE	Information Technology
46	Krushnaraj Jadhav	Information Technology
307347	Prasad Khadse	Information Technology
307351	Rushadi Gaidhane	Information Technology
64	Sarthak Raut	Information Technology
66	Suraj Zagare	Information Technology
30	Tejaswini Bhagat	Information Technology
307366	Vaishnavi Kaware	Information Technology

Copyright IIT Bombay 2024

34	vasudha kavishwar	Information Technology

	SHEGAON 444 203
Nodal Coordinator	Institute Stamp / Seal

1. Scan and Upload the duly signed ORIGINAL SOFT COPY of attendance.
2. Keep the HARD COPY with you in the Virtual Labs file for the record and the verification.

Copyright IIT Bombay 2024

7. Conclusion

The adoption of Virtual Labs as an innovative teaching-learning practice has significantly enriched the educational experience. It has provided students with an engaging, safe, and cost-effective platform to develop practical skills and reinforce theoretical knowledge. Moving forward, the integration of Virtual Labs with blended learning strategies can further enhance student learning outcomes and prepare them for technologically advanced work environments.

Prof. A G Sharma

Subject Incharge